WEB SERVERS IN JAVA:

OVERVIEW OF JAVA WEB SERVER, TOMCAT, WEBLOGIC

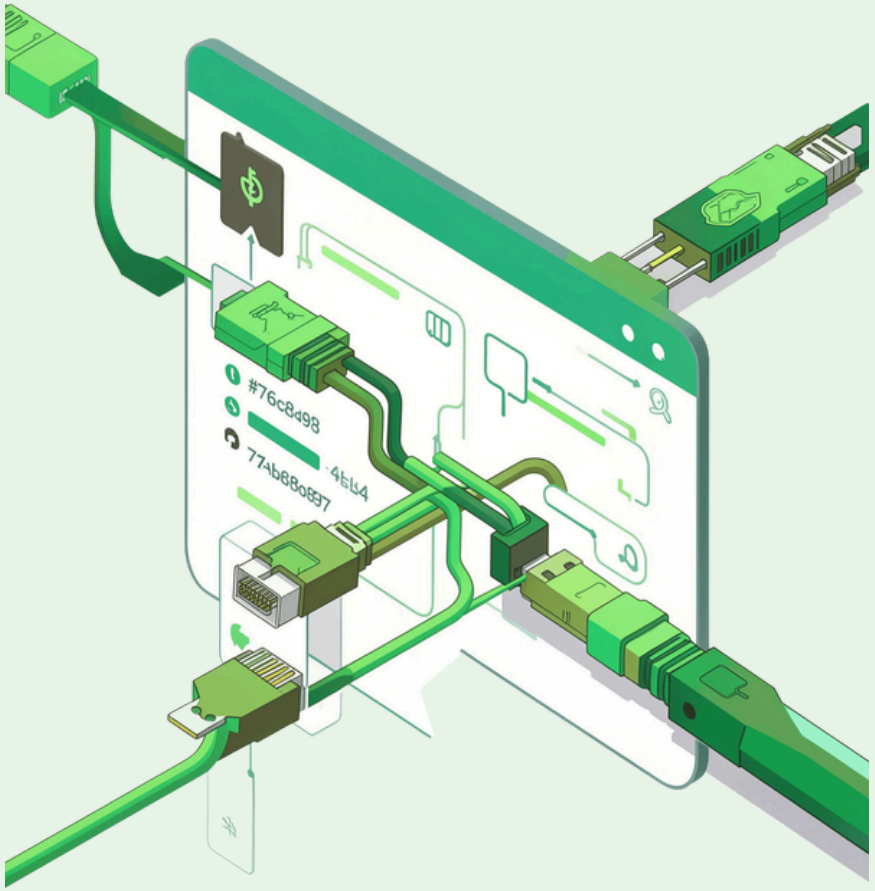
Java and Web Programming

NAME :- MD SAHID
ROLL NO. :- 26MCAB32



Web Servers in Java

A Java Web Server is software that receives requests from web browsers, runs Java web applications, and sends the required response back to the user.



What is a Java Web Server?

Handles HTTP Traffic

Receives HTTP requests from browsers and clients, then sends back appropriate responses — HTML, JSON, or other data.

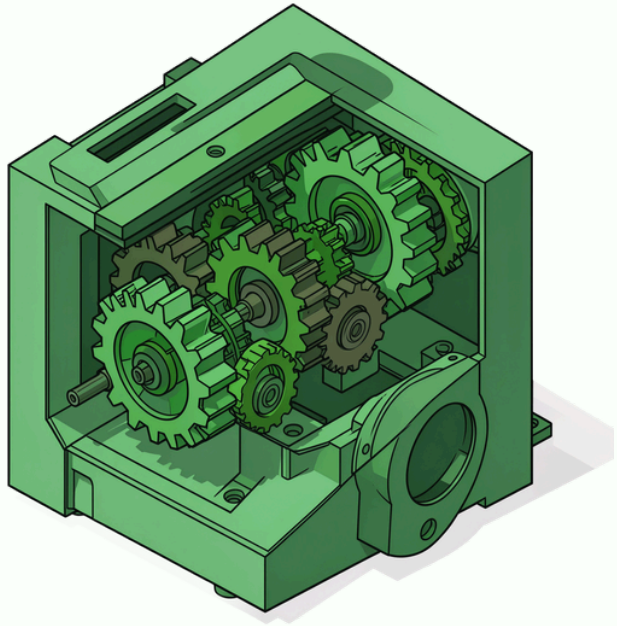
Executes Java Code

Runs Java servlets and JSP pages to generate dynamic, personalized web content on the fly.

Manages Connections

Handles hundreds or thousands of concurrent user sessions, allocating resources efficiently and reliably.

Key Responsibilities



Request Handling

Receives and processes HTTP requests from clients, routing them to the correct application logic.

Application Execution

Runs Java servlets and JSP pages, producing dynamic responses tailored to each request.

Resource Management

Efficiently manages memory, threads, and database connections to keep applications stable under load.



Apache Tomcat: The Lightweight Choice

Type

Open-source **servlet container** — not a full application server, but powerful for most web applications.

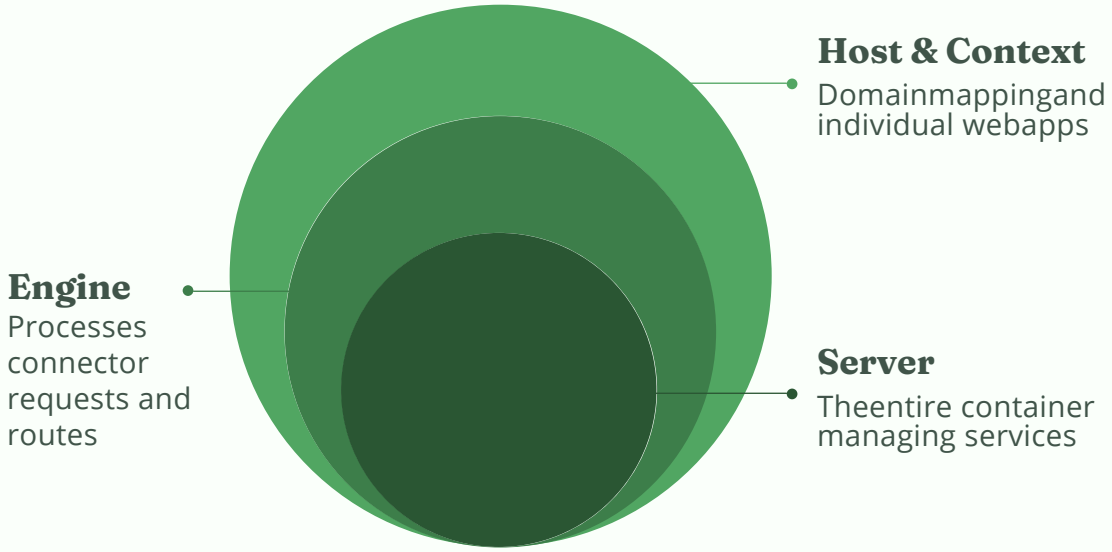
Maintained By

Apache Software **Foundation** — one of the world's most trusted open-source communities.

Best For

Straightforward web applications, **microservices**, and modern Spring Boot deployments.

Tomcat Architecture



How It All Fits Together

Tomcat's architecture is elegantly layered. At the top, the **Server** wraps everything. The **Engine** receives requests from connectors and routes them appropriately. Each **Host** maps a domain name to the server, and each **Context** represents a deployed web application — keeping concerns cleanly separated.

Tomcat Strengths



Lightweight & Fast

Minimal overhead means faster startup times and lower resource consumption — ideal for cloud environments.



Free & Open-Source

Zero licensing costs. Freely available, community-supported, and battle-tested across millions of deployments.



Easy Configuration

Plain text XML configuration files make setup and management straightforward, even for beginners.



Spring Boot Ready

Tomcat ships embedded inside Spring Boot — making it the default server for modern Java microservices and cloud deployments.

Oracle WebLogic: Enterprise-Grade Power

Type

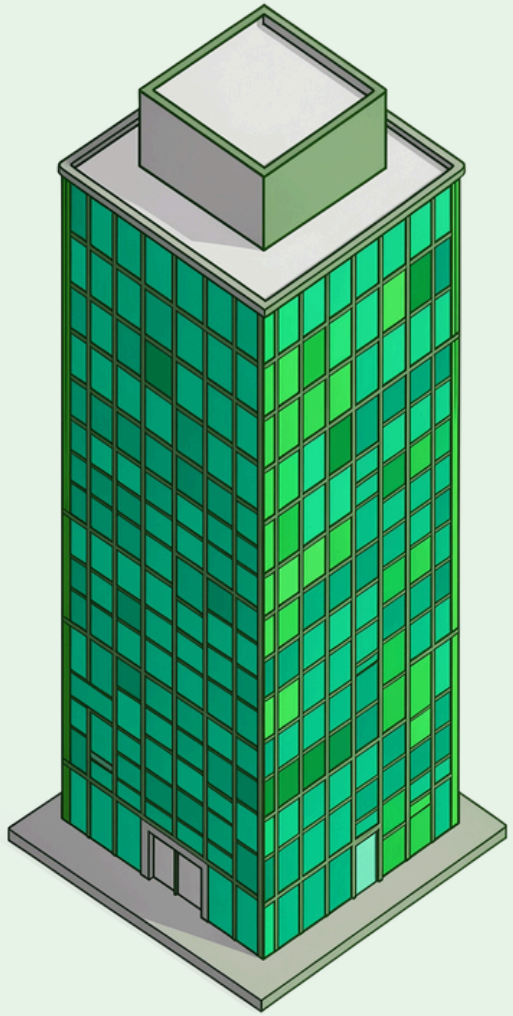
A **full-featured application server** implementing the complete Jakarta EE specification.

Maintained By

Oracle Corporation — with commercial licensing and dedicated enterprise support contracts.

Best For

Complex, **mission-critical enterprise applications** in finance, healthcare, and government sectors.



WebLogic Capabilities

Enterprise Features at a Glance

1

Full Jakarta EE Support

Enterprise JavaBeans (EJB), JMS messaging, and distributed transactions — the full enterprise stack.

2

Clustering & Failover

Built-in high availability, load balancing, and automatic failover to keep critical systems online.

3

Advanced Management

Rich monitoring dashboards, diagnostic tools, and administrative consoles for production operations.



WebLogic Strengths



Complete Jakarta EE Implementation

Implements the full Jakarta EE specification — every enterprise API and feature is available out of the box.



Enterprise-Grade Reliability

Clustering, distributed transactions, and JMS messaging make it the backbone of high-stakes systems.



Commercial Oracle Support

Backed by Oracle's global support network — critical for organizations that cannot afford downtime.



Proven in Critical Sectors

Trusted by financial institutions, healthcare providers, and government agencies where reliability is non-negotiable.

Tomcat vs. WebLogic: Quick Comparison

Category	Apache Tomcat	Oracle WebLogic
Licensing	Free & open-source	Commercial license required
Complexity	Simple, easy to configure	Feature-rich, steeper learning curve
Jakarta EE	Servlet/JSP only (partial)	Full specification support
Clustering	Limited, requires third-party tools	Built-in, enterprise-grade
Best Use Case	Microservices, Spring Boot apps	Mission-critical enterprise systems
Support	Community-driven	Oracle commercial support

Choosing the Right Server

Choose Apache Tomcat if...

- You are building modern web applications or microservices
- Your project uses **Spring Boot** — Tomcat is embedded by default
- You need a lightweight, cost-free, easy-to-deploy solution
- You are deploying to the cloud or containerized environments

Choose Oracle WebLogic if...

- You are running **complex enterprise systems** at scale
- Your application requires full Jakarta EE features: EJB, messaging, distributed transactions
- High availability, clustering, and failover are non-negotiable
- You operate in **finance, healthcare, or government** where Oracle support is essential

When in doubt, start with Tomcat — it handles the vast majority of Java web application needs with simplicity and speed. Graduate to WebLogic when enterprise complexity demands it.

CONCLUSION

- A JAVA WEB SERVER HELPS JAVA APPLICATIONS COMMUNICATE WITH USERS THROUGH THE WEB.
- IT RECEIVES HTTP REQUESTS, PROCESSES THEM, AND SENDS RESPONSES BACK TO THE USER.
- APACHE TOMCAT IS A SIMPLE, LIGHTWEIGHT, AND OPEN-SOURCE CHOICE FOR MANY JAVA WEB APPLICATIONS.
- ORACLE WEBLOGIC IS DESIGNED FOR LARGE AND COMPLEX ENTERPRISE APPLICATIONS.
- TOMCAT IS SUITABLE WHEN WE NEED SIMPLICITY AND EASY DEPLOYMENT.
- WEBLOGIC IS SUITABLE WHEN WE NEED ADVANCED FEATURES, HIGH AVAILABILITY, CLUSTERING, AND ENTERPRISE SUPPORT.

THANK YOU !!!